

Engineering Nonequilibrium Steady States via Quantum Optimal Control

Abdrakhman Akchurin^{*}, Alexander Schnell[†], and Weijian Chen^{*}

^{*} *North Carolina State University, United States*

[†] *Technical University of Berlin, Germany*

Abstract: Periodically driven open quantum systems can evolve, after initial transient dynamics, into nonequilibrium steady states (NESSs) in which the quantum state continues to evolve in time while the stroboscopic state observed at integer multiples of the drive period remains unchanged. In earlier work, we showed that Floquet engineering provides a flexible tool for stabilizing a single dissipative qubit at various high-purity NESSs. In the present study, we build on this result by combining Floquet engineering with quantum optimal control theory to systematically design time-periodic control protocols for dissipative qubit systems. We develop a gradient-based optimization framework that computes analytic derivatives of the Floquet propagator with respect to the control parameters, providing efficient gradients for maximizing NESS purity. Within this framework, we explore protocols that stabilize entangled steady states of two qubits in the presence of dissipation. These results illustrate how dissipation engineering, combined with Floquet-based optimal control, become a resource for robust quantum state preparation in quantum technologies.

Keywords: open quantum systems; Floquet engineering; quantum optimal control; nonequilibrium steady states; steady-state entanglement.